

DSA 8301: Statistical Inference for Big Data

Lecture Notes

Topic 1: Parametric Statistical Models

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1 Parametric Statistical Models

Statistical modeling provides a formal framework for describing, analyzing, and making inferences about real-world phenomena under uncertainty. Among the various modeling paradigms, **parametric statistical models** play a central role in classical statistics, econometrics, biostatistics, and many applied data science problems.

Parametric models assume that the data are generated from a probability distribution that belongs to a known family characterized by a *finite number of parameters*. This assumption allows for efficient estimation, hypothesis testing, and prediction, particularly when the sample size is moderate or small.

Despite their advantages, parametric models rely heavily on strong assumptions. Understanding their formulation, assumptions, and limitations is therefore essential for sound statistical practice.

1.1 Definition and Formulation of Parametric Statistical Models

1.1.1 Conceptual Introduction

In statistical inference, we aim to describe the mechanism that generates observable data using probability distributions. A **statistical model** provides a mathematical representation of this data-generating process.

A **parametric statistical model** assumes that the distribution of the data belongs to a known family of probability distributions characterized by a finite number of unknown parameters.

Definition 1.1. *Let:*

- $\mathcal{X}$ denote the **sample space**, i.e., the set of all possible observable outcomes.
- $\Theta \subseteq \mathbb{R}^p$ denote the **parameter space**.
- $\theta = (\theta_1, \theta_2, \dots, \theta_p)$ be a **parameter vector**.

A **parametric statistical model** is defined as a family of probability distributions

$$\mathcal{M} = \{f(x \mid \theta) : \theta \in \Theta\},$$

where $f(x \mid \theta)$ is either:

- a probability mass function (pmf), if X is discrete, or
- a probability density function (pdf), if X is continuous.

The defining feature of a parametric model is that the dimension p of the parameter vector is **finite and fixed**, and does not depend on the sample size n .

1.1.2 Key Components of a Parametric Model

A parametric model consists of the following fundamental elements:

1. **Sample Space ($\mathcal{X}$)** The set of all possible realizations of the random variable X .
Examples:

- $\mathcal{X} = \{0, 1\}$ for a Bernoulli experiment,
- $\mathcal{X} = \mathbb{R}$ for a continuous measurement.

2. **Parameter Space (Θ)** The set of all admissible parameter values. For example:

$$\Theta = (0, 1) \quad \text{for a Bernoulli model,}$$

a Bernoulli model with success probability θ , the parameter must lie between 0 and 1. Hence the parameter space is

$$\Theta = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\} \quad \text{for a Normal model.}$$

3. **Family of Distributions** Each value of θ specifies one probability distribution.

1.2 Common Families of Parametric Distributions

Parametric distributions provide mathematically tractable models for random phenomena by assuming that data arise from a probability distribution characterized by a finite number of parameters. These distributions form the building blocks of many statistical models used in estimation, hypothesis testing, and prediction.

1.2.1 Discrete Distributions

Discrete parametric distributions are used when the random variable takes values in a finite or countably infinite set, such as counts or categorical outcomes.

Bernoulli Distribution A random variable X follows a Bernoulli distribution with parameter $p \in (0, 1)$ if

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

The Bernoulli distribution models a single binary experiment with two possible outcomes, commonly interpreted as “success” ($x = 1$) and “failure” ($x = 0$).

Key properties:

- Mean: $\mathbb{E}[X] = p$
- Variance: $\text{Var}(X) = p(1 - p)$

The Bernoulli distribution is fundamental in statistics and serves as the basis for logistic regression and binary classification problems.

Binomial Distribution A random variable X follows a Binomial distribution with parameters $n \in \mathbb{N}$ and $p \in (0, 1)$ if

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

The Binomial distribution represents the number of successes in n independent Bernoulli trials, each with the same probability of success p .

Key properties:

- Mean: $\mathbb{E}[X] = np$
- Variance: $\text{Var}(X) = np(1 - p)$

The Binomial distribution is widely used in quality control, clinical trials, and survey sampling.

Poisson Distribution A random variable X is said to follow a Poisson distribution with rate parameter $\lambda > 0$ if

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

The Poisson distribution models the number of events occurring in a fixed interval of time or space, under the assumptions that events occur independently and at a constant average rate.

Key properties:

- Mean: $\mathbb{E}[X] = \lambda$
- Variance: $\text{Var}(X) = \lambda$

It is commonly applied in fields such as epidemiology, insurance, telecommunications, and traffic flow analysis.

1.2.2 Continuous Distributions

Continuous parametric distributions are used when the random variable can take any value in a continuous interval, typically subsets of the real line.

Normal Distribution A continuous random variable X follows a Normal (Gaussian) distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$ if its probability density function is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

The Normal distribution is symmetric about its mean and is fully characterized by its location parameter μ and scale parameter σ^2 .

Key properties:

- Mean: $\mathbb{E}[X] = \mu$
- Variance: $\text{Var}(X) = \sigma^2$

The Normal distribution plays a central role in statistics due to the Central Limit Theorem, which justifies its widespread use in modeling measurement errors and natural variability.

Exponential Distribution A random variable X follows an Exponential distribution with rate parameter $\lambda > 0$ if

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0.$$

The Exponential distribution is commonly used to model waiting times until the occurrence of an event.

Key properties:

- Mean: $\mathbb{E}[X] = \frac{1}{\lambda}$
- Variance: $\text{Var}(X) = \frac{1}{\lambda^2}$

A distinctive feature of the Exponential distribution is the **memoryless property**, which states that

$$P(X > s + t \mid X > s) = P(X > t), \quad s, t > 0.$$

Gamma Distribution A random variable X follows a Gamma distribution with shape parameter $\alpha > 0$ and scale parameter $\beta > 0$ if

$$f(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x > 0.$$

The Gamma distribution generalizes the Exponential distribution and provides flexibility in modeling positive, skewed data.

Key properties:

- Mean: $\mathbb{E}[X] = \alpha\beta$
- Variance: $\text{Var}(X) = \alpha\beta^2$

It is widely used in reliability analysis, survival analysis, and Bayesian statistics.

Special Cases of the Gamma Distribution

The Gamma distribution is a very flexible family that includes several important distributions as special cases obtained by particular choices of the shape parameter α and scale parameter β .

1. Exponential Distribution If $\alpha = 1$, the Gamma distribution reduces to the **Exponential distribution** with scale parameter β :

$$f(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x > 0.$$

Equivalently, using the rate parameter $\lambda = 1/\beta$,

$$f(x) = \lambda e^{-\lambda x}.$$

Properties:

- Mean: $\mathbb{E}[X] = \beta$
- Variance: $\text{Var}(X) = \beta^2$
- Memoryless property

Thus, the Exponential distribution is a special case:

$$\text{Gamma}(1, \beta) \equiv \text{Exponential}(\beta).$$

2. Chi-Square Distribution If $\alpha = \frac{\nu}{2}$ and $\beta = 2$, the Gamma distribution reduces to the **Chi-square distribution** with ν degrees of freedom:

$$\chi_\nu^2 \sim \text{Gamma}\left(\frac{\nu}{2}, 2\right).$$

Properties:

- Mean: ν
- Variance: 2ν

The chi-square distribution plays a central role in:

- Variance estimation,
- Hypothesis testing,
- Likelihood ratio tests.

3. Erlang Distribution If $\alpha = k$ is a positive integer and $\beta > 0$, the Gamma distribution becomes the **Erlang distribution**:

$$\text{Erlang}(k, \beta) = \text{Gamma}(k, \beta).$$

Interpretation:

- It represents the waiting time until the k -th event in a Poisson process.
- It is the sum of k independent $\text{Exponential}(\beta)$ variables.

4. Sum of Exponentials If

$$X_i \sim \text{Exponential}(\beta), \quad i = 1, \dots, k,$$

are independent, then

$$\sum_{i=1}^k X_i \sim \text{Gamma}(k, \beta).$$

Thus, the Gamma distribution generalizes the Exponential distribution to multiple waiting times.

5. Limiting Normal Approximation For large α , the Gamma distribution can be approximated by a Normal distribution via the Central Limit Theorem:

$$X \sim \text{Gamma}(\alpha, \beta) \Rightarrow X \approx N(\alpha\beta, \alpha\beta^2) \quad \text{for large } \alpha.$$

This explains why the Gamma distribution becomes more symmetric as α increases.

Summary of Important Special Cases

Condition	Resulting Distribution	Interpretation
$\alpha = 1$	Exponential	Waiting time to first event
$\alpha = k \in \mathbb{N}$	Erlang	Waiting time to k -th event
$\alpha = \nu/2, \beta = 2$	Chi-square	Sum of squared normals
Large α	Approx. Normal	CLT approximation

Conclusion: The Gamma distribution is a unifying family for modeling positive, right-skewed random variables and plays a fundamental role in probability theory, statistical inference, reliability analysis, and Bayesian modeling.

1.3 Exponential Family of Distributions

1.3.1 Motivation

Many commonly used probability distributions share a common mathematical structure that allows unified treatment of estimation, inference, and model construction. This class of distributions is known as the **exponential family**.

1.3.2 General Form

A parametric distribution belongs to the **exponential family** if its density or mass function can be written as

$$f(x \mid \theta) = h(x) \exp \left(\eta(\theta)^\top T(x) - A(\theta) \right),$$

where:

- $T(x)$ is the **sufficient statistic**,
- $\eta(\theta)$ is the **natural (canonical) parameter**,
- $A(\theta)$ is the **log-partition function**,
- $h(x)$ is the **base measure**.

1.3.3 Interpretation of Each Component

(a) Sufficient Statistic $T(x)$ The function $T(x)$ summarizes all information in the data relevant to inference about θ .

For independent observations $X_1, \dots, X_n$:

$$f(x_1, \dots, x_n \mid \theta) = \left[\prod_{i=1}^n h(x_i) \right] \exp \left(\eta(\theta)^\top \sum_{i=1}^n T(x_i) - nA(\theta) \right).$$

Thus,

$$\sum_{i=1}^n T(X_i)$$

is sufficient for θ .

Importantly, the dimension of $T(x)$ does **not** grow with sample size. This makes exponential families particularly powerful.

(b) Natural Parameter $\eta(\theta)$ The reparameterized form $\eta(\theta)$ is called the **canonical** or **natural parameter**.

In many cases, statistical theory becomes simpler when expressed in terms of η rather than θ .

(c) Log-Partition Function $A(\theta)$ The function $A(\theta)$ ensures that the density integrates to one:

$$\int h(x) \exp(\eta(\theta)^\top T(x) - A(\theta)) dx = 1.$$

Thus,

$$A(\theta) = \log \int h(x) \exp(\eta(\theta)^\top T(x)) dx.$$

It plays a crucial role in determining:

- The mean,
- The variance,
- The cumulants of the distribution.

1.3.4 Examples

(a) Bernoulli Distribution

$$f(x | \theta) = \theta^x (1 - \theta)^{1-x}.$$

Rewrite as:

$$f(x | \theta) = \exp \left(x \log \frac{\theta}{1 - \theta} + \log(1 - \theta) \right).$$

Here:

- $T(x) = x$,
- $\eta(\theta) = \log \frac{\theta}{1 - \theta}$,
- $A(\theta) = -\log(1 - \theta)$.

(b) Poisson Distribution

$$f(x \mid \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

Rewrite as:

$$f(x \mid \lambda) = \frac{1}{x!} \exp(x \log \lambda - \lambda).$$

Thus:

- $T(x) = x$,
- $\eta(\lambda) = \log \lambda$,
- $A(\lambda) = \lambda$.

(c) Normal Distribution (Known Variance)

$$f(x \mid \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

This can be written in exponential family form with:

- $T(x) = x$,
- $\eta(\mu) = \frac{\mu}{\sigma^2}$,
- $A(\mu) = \frac{\mu^2}{2\sigma^2}$.

1.3.5 Why the Exponential Family Is Important

1. **Fixed-Dimension Sufficiency** The sufficient statistic has fixed dimension regardless of n .
2. **Conjugate Priors in Bayesian Statistics** Exponential family likelihoods admit conjugate priors, leading to closed-form posterior distributions.
3. **Generalized Linear Models (GLMs)** GLMs assume that:
 - The response distribution belongs to the exponential family,
 - The mean is linked to predictors via a link function.

Examples:

- Logistic regression (Bernoulli),

- Poisson regression,
- Normal linear regression.

4. **Likelihood Simplicity** The log-likelihood has a simple additive structure:

$$\ell(\theta) = \eta(\theta)^\top \sum T(x_i) - nA(\theta).$$

This greatly simplifies maximum likelihood estimation.

1.4 Exponential Family of Distributions

A parametric distribution belongs to the **exponential family** if its density or mass function can be expressed in the form

$$f(x \mid \theta) = h(x) \exp \left(\eta(\theta)^\top T(x) - A(\theta) \right),$$

where:

- $T(x)$ is a sufficient statistic,
- $\eta(\theta)$ is the natural (canonical) parameter,
- $A(\theta)$ is the log-partition function ensuring normalization.

The exponential family is particularly important because it admits sufficient statistics of fixed dimension and leads to simplified likelihood-based inference.

Distributions such as the Bernoulli, Binomial, Poisson, Normal, and Gamma all belong to this family. Moreover, the exponential family provides the theoretical foundation for **generalized linear models (GLMs)**, which unify regression modeling for diverse types of response variables.

1.5 Model Assumptions and Identifiability in Parametric Statistical Models

1.5.1 Introduction

In parametric statistical modeling, we assume that observed data are generated from a probability distribution belonging to a family indexed by a finite number of parameters. Formally, a parametric statistical model is defined as

$$\mathcal{M} = \{f(x|\theta) : \theta \in \Theta\},$$

where $f(x|\theta)$ denotes the probability density function (or mass function) of the data and θ is an unknown parameter vector.

For statistical inference to be meaningful, the model must satisfy certain assumptions and the parameters must be **identifiable**. If these conditions are violated, reliable estimation and inference become impossible.

1.5.2 Model Assumptions

Model assumptions specify the conditions under which the statistical model accurately represents the data-generating mechanism.

Distributional Assumption

The primary assumption is that the data follow a specific probability distribution from the parametric family.

$$X \sim f(x|\theta)$$

For example:

- Normal model: $X \sim N(\mu, \sigma^2)$
- Poisson model: $X \sim \text{Poisson}(\lambda)$
- Bernoulli model: $X \sim \text{Bernoulli}(p)$

If the chosen distribution does not reflect the true data-generating process, the model becomes **misspecified**.

Independence Assumption

Often we assume observations are independent:

$$X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} f(x|\theta)$$

Independence simplifies the likelihood function:

$$L(\theta|x_1, \dots, x_n) = \prod_{i=1}^n f(x_i|\theta).$$

Violations of independence commonly occur in:

- time series data
- spatial data
- clustered observations

Homogeneity (Stationarity) Assumption

The model often assumes that all observations share the same parameter θ .

$$X_i \sim f(x|\theta), \quad i = 1, \dots, n.$$

This assumption implies that the data are generated under identical conditions. If parameters change over time or across groups, the model may require extensions such as hierarchical or mixture models.

Regularity Conditions

For theoretical results (such as maximum likelihood estimation) to hold, certain regularity conditions are assumed:

- The parameter space Θ is open.
- The likelihood function is differentiable.
- Expectations and derivatives can be interchanged.

These conditions ensure desirable statistical properties such as consistency and asymptotic normality of estimators.

1.5.3 Identifiability of Parameters

A parametric model is said to be **identifiable** if different parameter values correspond to different probability distributions.

Definition 1.2. *A model is identifiable if*

$$f(x|\theta_1) = f(x|\theta_2)$$

for all x implies

$$\theta_1 = \theta_2.$$

This means that the parameter value can be uniquely determined from the distribution of the observed data.

Importance of Identifiability

Identifiability is essential because:

- Parameters cannot be estimated uniquely without it.
- Maximum likelihood estimators may not exist or may be non-unique.

- Bayesian posterior distributions may become ill-defined.

Thus identifiability is a fundamental requirement for valid statistical inference.

1.5.4 Examples of Identifiable Models

Normal Distribution

Consider the normal distribution

$$X \sim N(\mu, \sigma^2).$$

If

$$N(\mu_1, \sigma_1^2) = N(\mu_2, \sigma_2^2),$$

then

$$\mu_1 = \mu_2, \quad \sigma_1^2 = \sigma_2^2.$$

Hence the normal distribution is identifiable.

Bernoulli Distribution

For

$$X \sim \text{Bernoulli}(p),$$

the parameter p uniquely determines the distribution, so the model is identifiable.

1.5.5 Non-Identifiable Models

A model is **non-identifiable** if different parameter values produce the same distribution.

Example: Overparameterized Model

Consider the model

$$X \sim N(\theta_1 + \theta_2, 1).$$

Here only the sum $\theta_1 + \theta_2$ determines the distribution.

Thus many parameter combinations produce the same distribution:

$$(\theta_1, \theta_2) = (2, 1), (1, 2), (3, 0).$$

Hence the parameters are not uniquely identifiable.

Example: Mixture Models

Finite mixture models may also suffer from identifiability problems.

For example,

$$f(x) = \pi f_1(x|\theta_1) + (1 - \pi)f_2(x|\theta_2).$$

Swapping the components produces the same distribution:

$$(\pi, \theta_1, \theta_2) \leftrightarrow (1 - \pi, \theta_2, \theta_1).$$

Thus additional constraints are often required.

1.5.6 Consequences of Non-Identifiability

If a model is not identifiable:

- Parameter estimates may not be unique.
- The likelihood function may contain flat regions.
- Confidence intervals may become unbounded.
- Interpretation of parameters becomes ambiguous.

Therefore identifiability must be verified when constructing statistical models.

1.5.7 Consequences of Non-Identifiability

When a statistical model is **non-identifiable**, different parameter values correspond to the same probability distribution. As a result, the observed data cannot uniquely determine the parameter values. This leads to several practical and theoretical difficulties in statistical inference.

1. Parameter Estimates May Not Be Unique If the model is non-identifiable, several parameter values may produce exactly the same likelihood. Consequently, the estimator (such as the maximum likelihood estimator) is not unique.

Example.

Consider the model

$$X \sim N(\theta_1 + \theta_2, 1).$$

The likelihood depends only on the sum $\theta_1 + \theta_2$. Thus, many parameter pairs produce the same distribution:

$$(\theta_1, \theta_2) = (2, 1), (1, 2), (0, 3).$$

All these parameter combinations generate the same likelihood. Therefore the parameters θ_1 and θ_2 cannot be uniquely estimated.

Illustration.

The likelihood function depends only on

$$\mu = \theta_1 + \theta_2.$$

Hence the likelihood is constant along lines

$$\theta_1 + \theta_2 = c.$$

Graphically, the likelihood surface forms a ridge rather than a single peak.

2. The Likelihood Function May Contain Flat Regions Non-identifiability often leads to likelihood functions that are flat along certain directions in the parameter space. This means that many parameter values give exactly the same likelihood.

Example.

Suppose we observe $X_1, \dots, X_n \sim N(\theta_1 + \theta_2, 1)$. The log-likelihood becomes

$$\ell(\theta_1, \theta_2) = -\frac{1}{2} \sum_{i=1}^n (x_i - (\theta_1 + \theta_2))^2 + C.$$

Since the likelihood depends only on $\theta_1 + \theta_2$, any pair (θ_1, θ_2) with the same sum produces the same value of the likelihood.

Thus the likelihood surface has a **ridge** instead of a single maximum.

Flat likelihood regions make optimization difficult and can cause numerical instability in estimation algorithms.

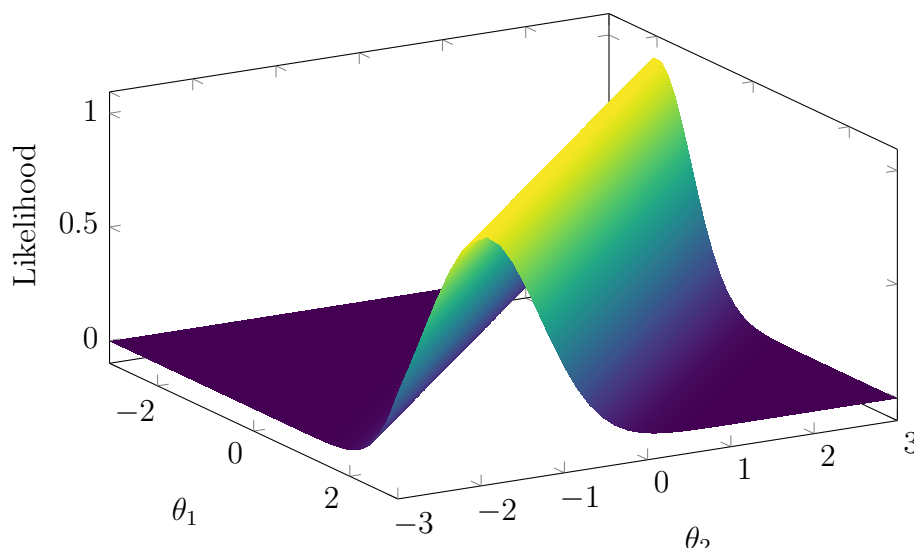


Figure 1: Example of a likelihood surface with a ridge. The likelihood depends only on $\theta_1 + \theta_2$, producing infinitely many parameter combinations with the same maximum likelihood. This illustrates non-identifiability.

3. Confidence Intervals May Become Unbounded When parameters are not identifiable, the data do not provide enough information to estimate them precisely. As a result, standard errors become very large and confidence intervals may become extremely wide or even unbounded.

Example.

Consider again the model

$$X \sim N(\theta_1 + \theta_2, 1).$$

While the sum $\theta_1 + \theta_2$ can be estimated, the individual parameters θ_1 and θ_2 cannot. Consequently, confidence intervals for these parameters cannot be meaningfully constructed.

4. Interpretation of Parameters Becomes Ambiguous If multiple parameter values correspond to the same distribution, then the parameters lose their interpretability.

Example: Mixture Models

Consider a two-component mixture model:

$$f(x) = \pi f_1(x|\theta_1) + (1 - \pi)f_2(x|\theta_2).$$

The model remains unchanged if we swap the components:

$$(\pi, \theta_1, \theta_2) \longleftrightarrow (1 - \pi, \theta_2, \theta_1).$$

Thus the components are not uniquely labeled. This is known as **label switching** in mixture models.

Without additional constraints, interpreting θ_1 as the “first component” becomes meaningless.

1.5.8 Strategies to Ensure Identifiability

Several approaches can be used to address identifiability issues:

- Reduce the number of parameters.
- Introduce constraints on parameters.
- Reparameterize the model.
- Use informative priors in Bayesian analysis.

Careful examination of assumptions and identifiability is therefore a crucial step in statistical modeling and data analysis.

Try Your Knowledge 1.1

Practical Exercises

1. Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with density $f(x|\theta)$.
 - (a) Define a **parametric statistical model**.
 - (b) Identify and explain the roles of the following components:
 - sample space $\mathcal{X}$,
 - parameter space Θ ,
 - family of distributions $\{f(x|\theta) : \theta \in \Theta\}$.
 - (c) Give two examples of parametric statistical models used in practice.
2. Suppose $X \sim \text{Bernoulli}(p)$.
 - (a) Write down the probability mass function of X .
 - (b) Specify the sample space and parameter space.
 - (c) Compute the mean and variance of X .
 - (d) Explain how the Bernoulli model can be used in binary classification problems.
3. Let $X \sim \text{Binomial}(n, p)$.
 - (a) Write down the probability mass function.
 - (b) Derive the mean and variance of X .
 - (c) Explain the relationship between the Binomial and Bernoulli distributions.
 - (d) Give one practical example where a Binomial model may be appropriate.

4. Suppose $X \sim \text{Poisson}(\lambda)$.
- (a) Write the probability mass function.
 - (b) Show that $\mathbb{E}[X] = \lambda$.
 - (c) Show that $\text{Var}(X) = \lambda$.
 - (d) Explain why the Poisson distribution is often used to model counts of rare events.

5. Let $X \sim N(\mu, \sigma^2)$.
- (a) Write the probability density function of the Normal distribution.
 - (b) Define the parameter space for (μ, σ^2) .
 - (c) State two important properties of the Normal distribution.
 - (d) Explain why the Normal distribution plays a central role in statistical inference.

6. Consider the exponential distribution with density

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0.$$

- (a) Derive the mean and variance of X .
- (b) Show that the exponential distribution satisfies the **memoryless property**

$$P(X > s + t | X > s) = P(X > t).$$

- (c) Give two real-world examples where the exponential distribution is appropriate.

7. Let $X \sim \text{Gamma}(\alpha, \beta)$ with density

$$f(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}.$$

- (a) Derive the mean and variance of the Gamma distribution.
- (b) Show that the exponential distribution is a special case of the Gamma distribution.
- (c) Explain the relationship between the Gamma and Chi-square distributions.

8. A probability distribution belongs to the exponential family if its density can be written as

$$f(x|\theta) = h(x) \exp(\eta(\theta)^\top T(x) - A(\theta)).$$

- (a) Explain the meaning of the functions $T(x)$, $\eta(\theta)$, and $A(\theta)$.

- (b) Show that the Poisson distribution belongs to the exponential family.
 - (c) Identify the sufficient statistic for the Poisson distribution.
9. Explain the concept of **identifiability** in parametric statistical models.

- (a) State the formal definition of identifiability.
- (b) Show that the model $X \sim N(\mu, \sigma^2)$ is identifiable.
- (c) Consider the model

$$X \sim N(\theta_1 + \theta_2, 1).$$

Explain why this model is not identifiable.

10. Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 .

- (a) Define the **sampling distribution** of an estimator.
- (b) Show that the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is an unbiased estimator of μ .

- (c) Derive the variance of $\bar{X}$.
- (d) Explain the role of the Central Limit Theorem in the sampling distribution of $\bar{X}$.

11. Let $X_1, \dots, X_n$ be a random sample from an exponential distribution with parameter λ .

- (a) Write the first population moment of the distribution.
- (b) Use the **method of moments** to derive an estimator for λ .
- (c) Discuss one advantage and one limitation of the method of moments.